FYUGP B.Sc. MATHEMATICS HONOURS
FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION
MATHEMATICS
MAT5CJ302 Abstract Algebra I
Model Question Paper

Maximum Time : 2 Hours	Maximum Marks : 70

Section A

All Questions can be answered. Each Question carries 3 marks (Ceiling : 24 Marks)

1	Let $*$ be defined on $\mathbb{Z}^+$ by $a * b = \min\{a, b\}$. Is $*$ a binary operation on $\mathbb{Z}^+$? Is it commutative? Justify your answer.	3
2	Let $*$ be defined on $\mathbb{Z}$ by $a * b = a - b$. Check whether $*$ is associative.	3
3	Show that the identity element of a set with a binary operation, if it exists, is unique.	3
4	Give an example of a set with a binary operation which is commutative but not associative.	3
5	Show that $\langle \mathbb{Q}^+, * \rangle$ with $a * b = \frac{ab}{3}$ is a group.	3
6	Show that $\langle \mathbb{Z}, + \rangle$ is an abelian group. Is $\langle \mathbb{Z}^+, + \rangle$ a group? Justify.	3
7	In a group G, show that $(a^{-1})^{-1} = a$ for every $a \in G$.	3
8	Write the group table of $\{1, i, -1, -i\}$ under multiplication.	3
9	Using only the group axioms, show that if $a * b = a * c$ in a group then $b = c$.	3
10	Show that $f : \mathbb{Z} \to 3\mathbb{Z}$ given by $f(m) = 3m$ is an isomorphism from $\langle \mathbb{Z}, + \rangle$ onto $\langle 3\mathbb{Z}, + \rangle$.	3

Section B		
All Questions can be answered. Each Question carries 6 marks (Ceiling : 36 Marks)		
11	Let $S = \mathbb{R} \setminus \{-1\}$ and define $*$ on S by $a * b = a + b + ab$. Show that $\langle S, * \rangle$ is a group.	6
12	Let G be a group and $a, b \in G$. Show that $(ab)^{-1} = b^{-1}a^{-1}$.	6
13	Show that $\{1, 2, 3, 4, 5, 6\}$ under multiplication modulo 7 is a group.	6
14	Construct the group table of $\mathbb{Z}_4$ and verify that each element appears exactly once in each row and each column.	6
15	Let G be a group. Prove that G is abelian if and only if $(ab)^2 = a^2b^2$ for all $a, b \in G$.	6
16	Let $f : G_1 \to G_2$ be an isomorphism of groups. Prove that $f(e_1) = e_2$, where e_1 and e_2 are the identities of G_1 and G_2.	6

Section C		
Answer any ONE. Each Question carries 10 marks (1 × 10 = 10 Marks)		
17	Let $*$ be defined on $\mathbb{R}$ by $a * b = \sqrt[3]{a^3 + b^3}$. Show that $\langle \mathbb{R}, * \rangle$ is an abelian group.	10
18	Let G be a group. Prove that the identity element of G is unique and that every element of G has a unique inverse, using only the group axioms.	10